CONTROLLER — SEVEN READABLE KICAD SHEETS

01 Power and Pi

File: 01_power_and_pi.kicad_sch

02 ESP32 and Pi UART

File: 02_esp32_and_pi.kicad_sch

03 Left Motor

File: 03_left_motor.kicad_sch

04 Right Motor

File: 04_right_motor.kicad_sch

05 Encoders

File: 05_encoder_inputs.kicad_sch

06 Arm Servos

File: 06_arm_servos.kicad_sch

07 Main Servo 8.4V

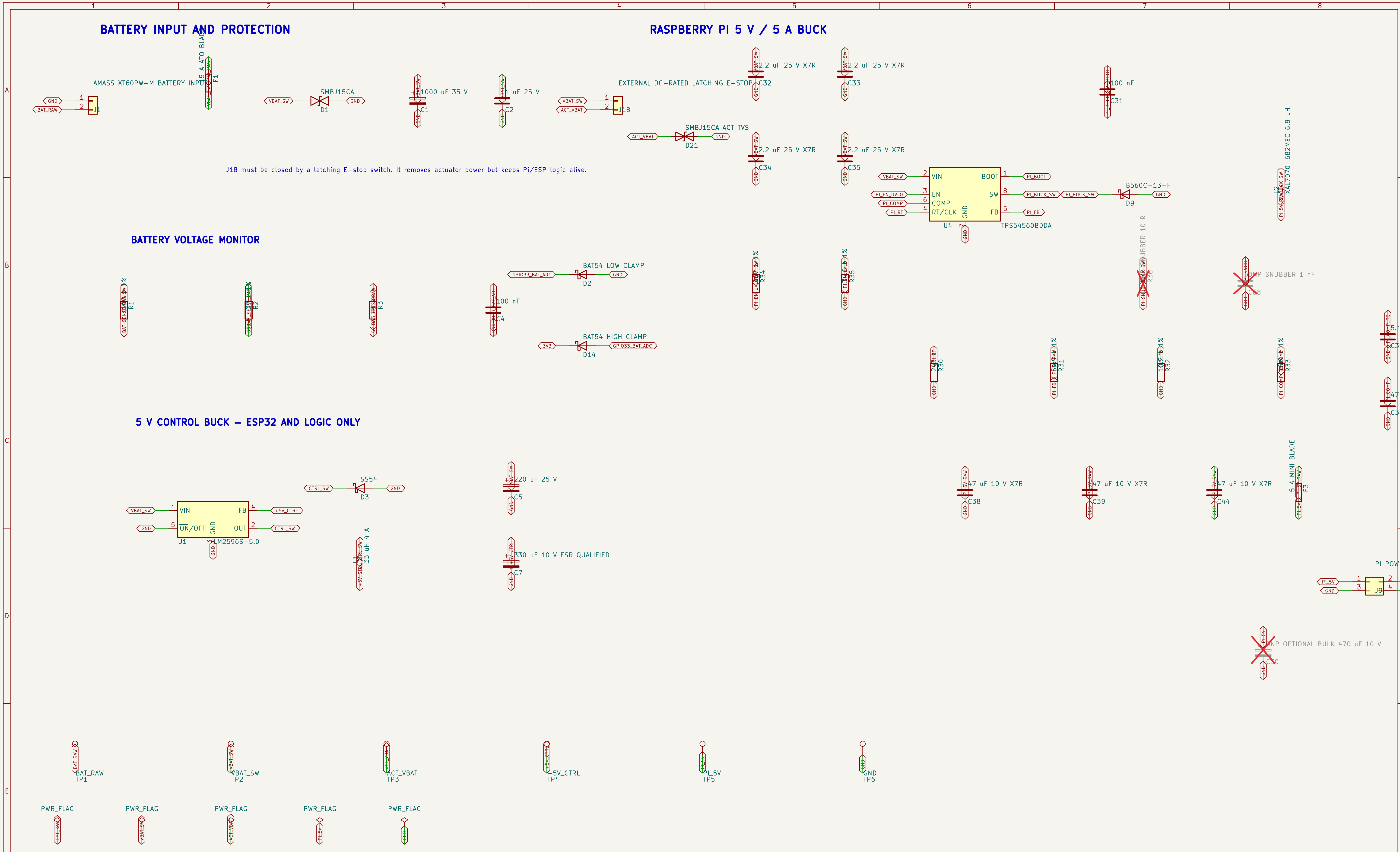
Harrison

Sheet: /

File: pickleball_robot_controller.kicad_sch

Title: Pickleball Robot Controller — Hierarchical Overview

Size: A4	Date: 2026-07-12	**Rev: R2-MULTISHEET**
KiCad E.D.A. 10.0.4		Id: 1/8



Sheet 1 of 7

Harrison

Sheet: /01 Power and Pi/

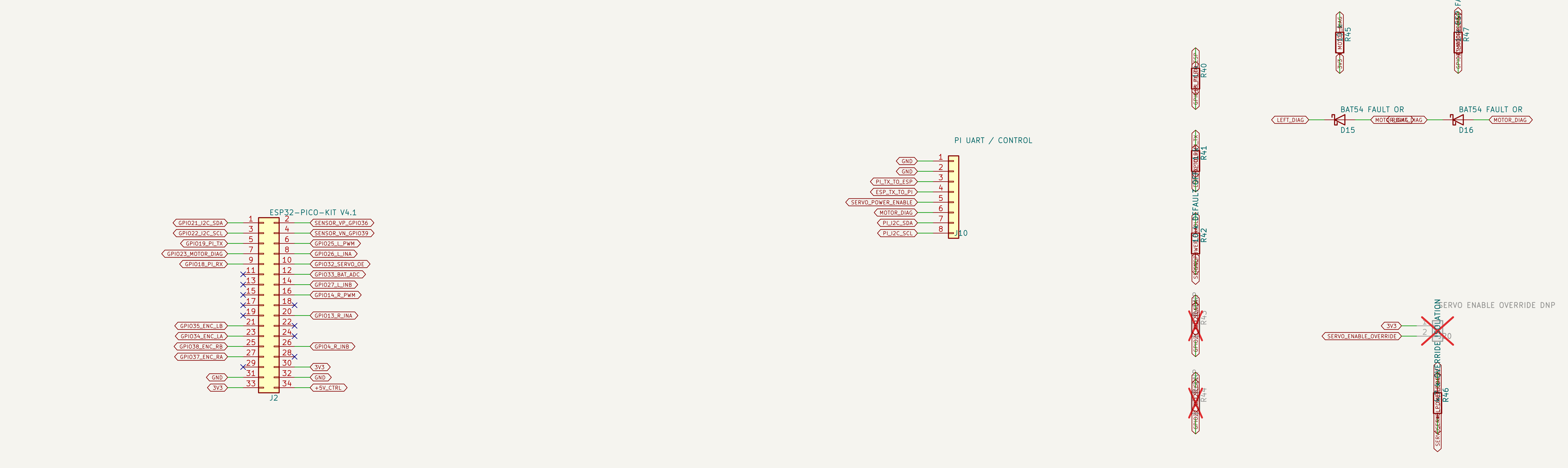
File: 01_power_and_pi.kicad_sch

Title: Battery Protection, Logic Buck, and Raspberry Pi Power

Size: A3	Date: 2026-07-12	Rev: R2-MULTISHEET
KiCad E.D.A. 10.0.4		Id: 2/8

ESP32-PICO-KIT V4.1 CARRIER — CONFIRMED 17.78 mm ROW SPACING

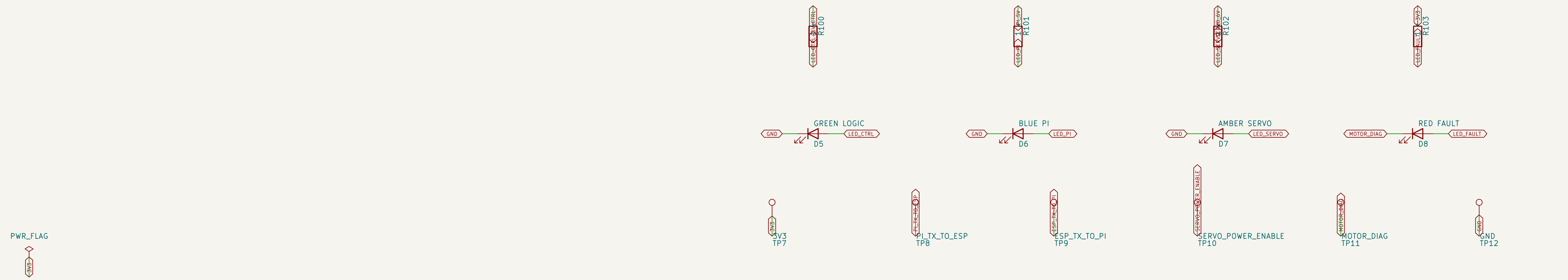
RASPBERRY PI UART / CONTROL — NORMAL 2.54 mm HEADER



J10: 1/2 GND; 3 Pi TX->ESP GPIO18; 4 ESP GPIO19->Pi RX; 5 servo-power enable; 6 motor fault; 7/8 I2C DNP.

Pi: GPIO14 pin8 -> J10.3; GPIO15 pin10 <- J10.4; GPIO17 pin11 -> J10.5; GPIO27 pin13 <- J10.6; GND -> J10.1/2.

STATUS LEDS

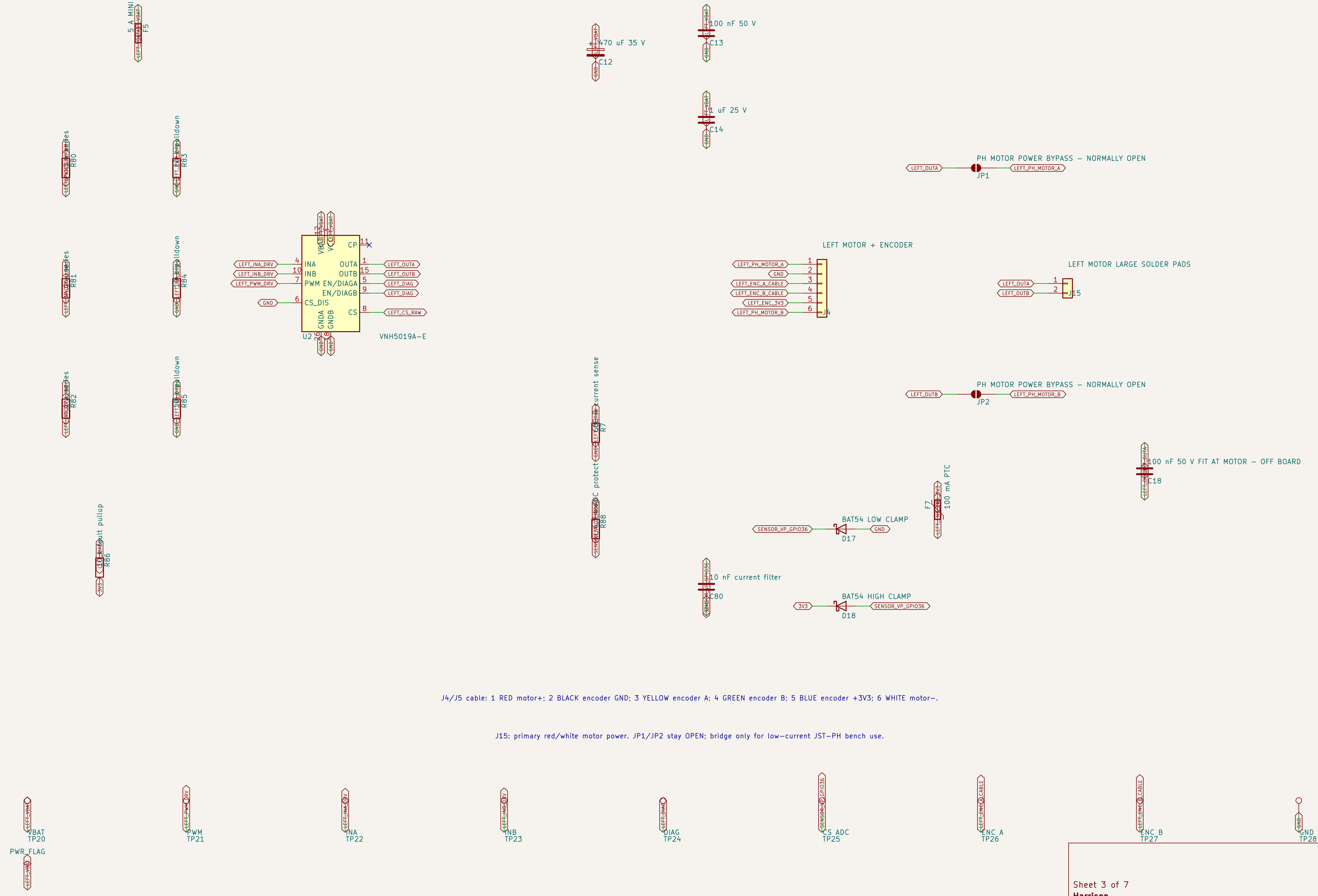


Sheet 2 of 7
Harrison
Sheet: /02 ESP32 and Pi UART/
File: 02_esp32_and_pi.kicad_sch

Title: ESP32 Carrier, Raspberry Pi UART, and Status LEDs

Size: A3	Date: 2026-07-12	**Rev: R2-MULTISHEET**
KiCad E.D.A. 10.0.4		Id: 3/8

LEFT WHEEL MOTOR DRIVER

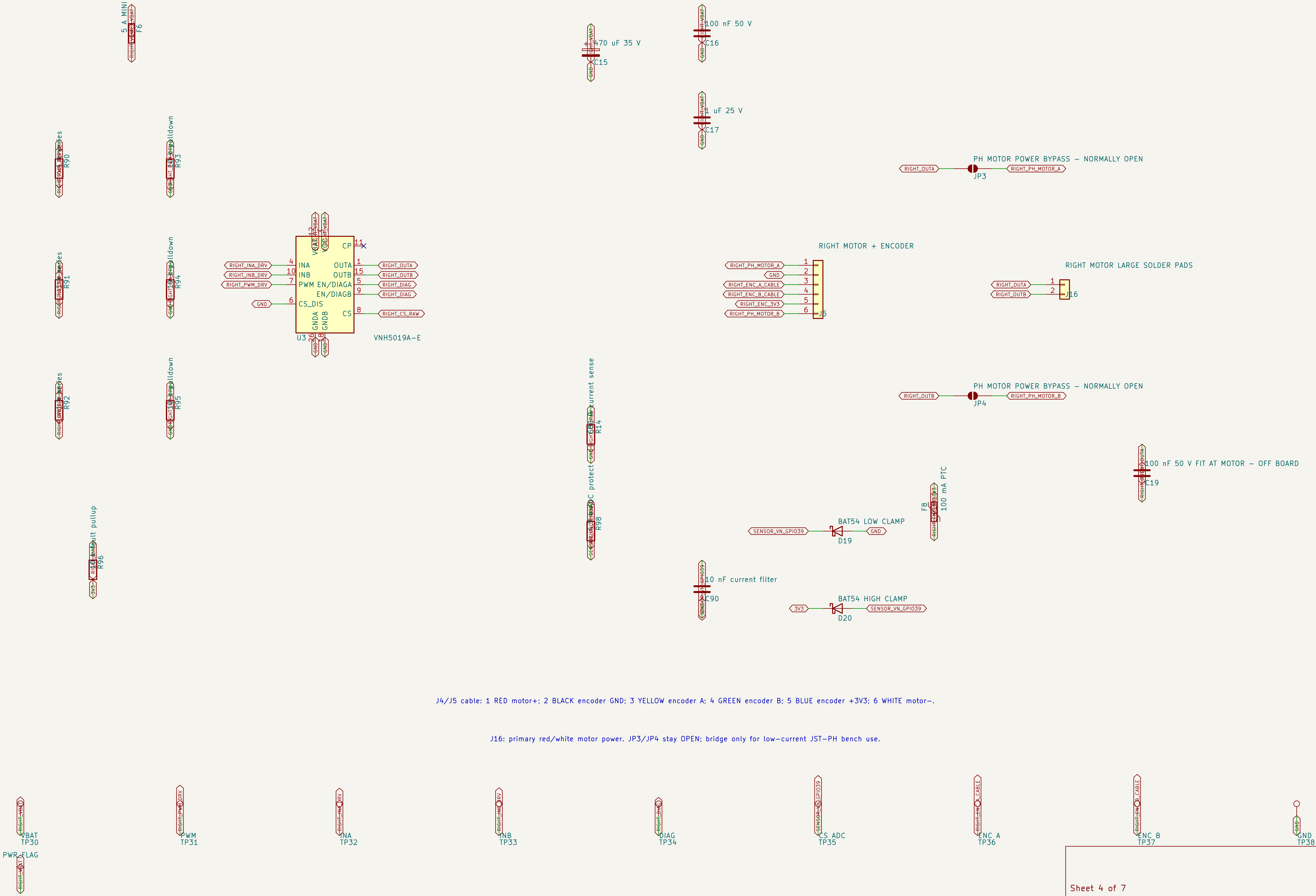


J4/J5 cable: 1 RED motor+; 2 BLACK encoder GND; 3 YELLOW encoder A; 4 GREEN encoder B; 5 BLUE encoder +3V3; 6 WHITE motor−.

J15: primary red/white motor power. JP1/JP2 stay OPEN; bridge only for low-current JST-PH bench use.

Sheet 3 of 7
Harrison
Sheet: /03 Left Motor/
File: 03_left_motor.kicad_sch
Title: Left Wheel Motor Driver
Size: A3 Date: 2026-07-12 **Rev: R2-MULTISHEET**
KiCad E.D.A. 10.0.4 Id: 4/8

RIGHT WHEEL MOTOR DRIVER



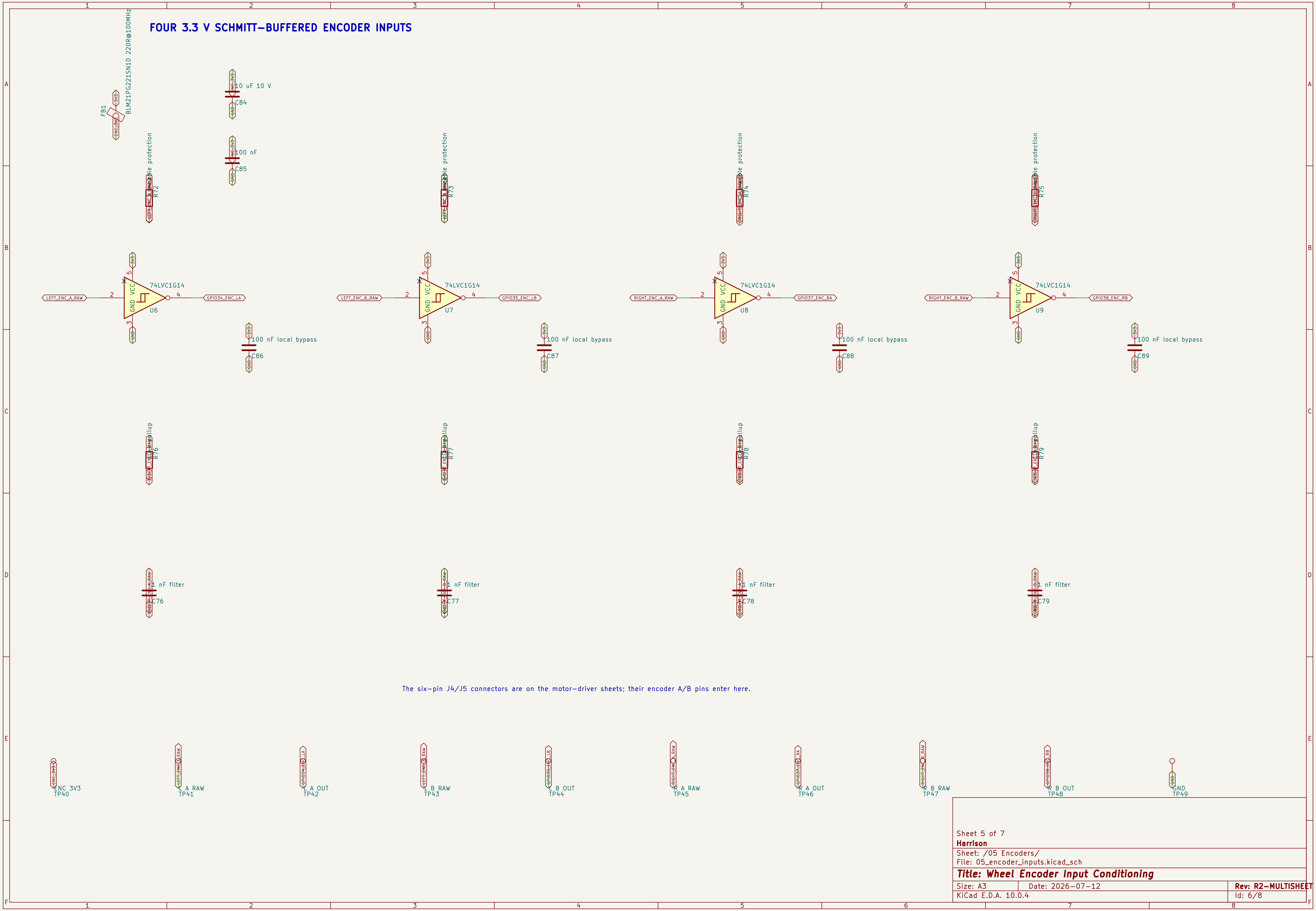
J4/J5 cable: 1 RED motor+; 2 BLACK encoder GND; 3 YELLOW encoder A; 4 GREEN encoder B; 5 BLUE encoder +3V3; 6 WHITE motor−.

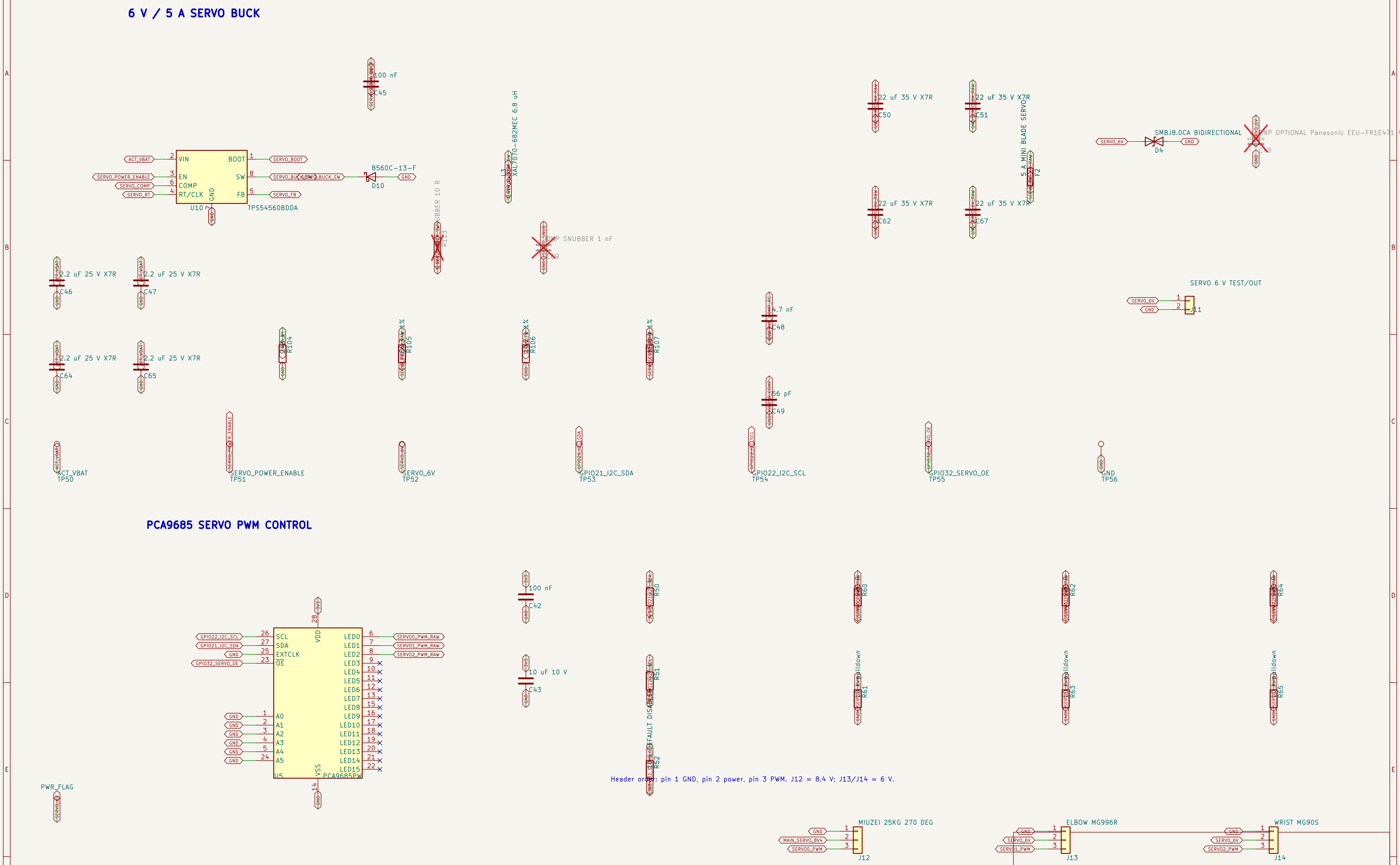
J16: primary red/white motor power. JP3/JP4 stay OPEN; bridge only for low-current JST-PH bench use.

Sheet 4 of 7
Harrison
Sheet: /04 Right Motor/
File: 04_right_motor.kicad_sch

Title: Right Wheel Motor Driver

Size: A3 | Date: 2026-07-12 | **Rev: R2-MULTISHEET**
KiCad E.D.A. 10.0.4 | Id: 5/8


FOUR 3.3 V SCHMITT–BUFFERED ENCODER INPUTS
FB1
BLM21PG221SN1D 220R@100MHz
10 uF 10 V
C84
100 nF
C85
R72
R73
R74
R75
74LVC1G14
U6
U7
U8
U9
LEFT_ENC_A_RAW
LEFT_ENC_B_RAW
RIGHT_ENC_A_RAW
RIGHT_ENC_B_RAW
GPIO34_ENC_LA
GPIO35_ENC_LB
GPIO37_ENC_RA
GPIO38_ENC_RB
100 nF local bypass
C86
C87
C88
C89
R76
R77
R78
R79
1 nF filter
C76
C77
C78
C79
The six–pin J4/J5 connectors are on the motor–driver sheets; their encoder A/B pins enter here.
ENC 3V3
TP40
L A RAW
TP41
L A OUT
TP42
L B RAW
TP43
L B OUT
TP44
R A RAW
TP45
R A OUT
TP46
R B RAW
TP47
R B OUT
TP48
GND
TP49
Sheet 5 of 7
Harrison
Sheet: /05 Encoders/
File: 05_encoder_inputs.kicad_sch
Title: Wheel Encoder Input Conditioning
Size: A3
Date: 2026–07–12
Rev: R2–MULTISHEET
KiCad E.D.A. 10.0.4
Id: 6/8



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Harrison

Sheet: /06 Arm Servos/

File: 06_arm_servos.kicad_sch

Title: Servo Power Regulator and 3-DOF Arm Control

Size: A3 | Date: 2026-07-12 | Rev: R2-MULTISHEET

KiCad E.D.A. 10.0.4 | Id: 7/8

DEDICATED 8.4 V / 5 A BUCK FOR MIUZEI 25KG 270 DEG SERVO

U11 uses 96.9 k / 10.2 k feedback resistors for approximately 8.40 V. J12 is the only servo powered from this rail.

J13 MG996R and J14 MG90S remain on the separate regulated 6 V rail.

Sheet 7 of 7
Harrison
Sheet: /07 Main Servo 8.4V/
File: 07_main_servo_8v4.kicad_sch

Title: Miuzei Main-Swing Servo 8.4 V Power Rail

Size: A3 | Date: 2026-07-12 | Rev: R2-MULTISHEET
KiCad E.D.A. 10.0.4 | Id: 8/8